

Time Series Exercise Sheet 3

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Exercise 3.1

Find the coefficients ψ_j $j = 0, 1, 2, 3, \dots$ in the representation of

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

of the ARMA process

$$(1 - 0.5B + 0.04B^2)X_t = (1 + 0.25B)\varepsilon_t$$

when ε_t is white noise.

Solution 3.1

We have

$$\begin{aligned} X_t &= \frac{1 + 0.25B}{1 - 0.5B + 0.04B^2} \varepsilon_t \\ &= \frac{1 + 0.25B}{0.04(B - 2.5)(B - 10)} \varepsilon_t \\ &= \frac{1 + 0.25B}{0.04 \times 7.5} \left(\frac{-1}{B - 2.5} + \frac{1}{B - 10} \right) \varepsilon_t \\ &= \frac{1 + 0.25B}{0.3} \left(\frac{1}{2.5} \sum_{j=0}^{\infty} 0.4^j B^j - \frac{1}{10} \sum_{j=0}^{\infty} 0.1^j B^j \right) \varepsilon_t \\ &= \sum_{j=0}^{\infty} \left\{ \left(\frac{0.4^j}{0.75} - \frac{0.1^j}{3} \right) B^j + \left(\frac{0.4^j}{3} - \frac{0.1^j}{12} \right) B^{j+1} \right\} \varepsilon_j \\ &= \sum_{j=0}^{\infty} \left(\frac{0.4^j}{0.75} - \frac{0.1^j}{3} \right) \varepsilon_{t-j} + \sum_{l=1}^{\infty} \left(\frac{0.4^l}{1.2} - \frac{0.1^l}{1.2} \right) \varepsilon_{t-l} \\ &= \varepsilon_t + \sum_{j=1}^{\infty} \left(\frac{0.4^j}{0.75} - \frac{0.1^j}{3} + \frac{0.4^j}{1.2} - \frac{0.1^j}{1.2} \right) \varepsilon_{t-j} \\ &= \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \end{aligned}$$

where

$$\psi_j = \begin{cases} 1 & \text{if } j = 0 \\ \frac{13}{6} 0.4^j - \frac{7}{6} 0.1^j & \text{if } j = 1, 2, \dots \end{cases}$$

Exercise 3.2

Assume that Y_t is a causal and invertible ARMA process $\phi(B)Y_t = \theta(B)\epsilon_t$ define $\tilde{\phi}(B) = \phi(B)\theta^{-1}(B)$ and take $a(B) = (\tilde{\phi}(B))^{-1} = \sum_{j=0}^{\infty} a_j B^j$. Determine the representation of Y_t in terms of ϵ_t , firstly when Y_t is a stationary AR(1) and secondly when Y_t is a stationary AR(2).

Solution 3.2

We start with the AR(1) model $(1 - \phi B)Y_t = \epsilon_t$. By causality, we may express Y_t as

$$Y_t = \frac{1}{1 - \phi B} \epsilon_t = \sum_{k=0}^{\infty} \phi^k \epsilon_{t-k}.$$

The AR(2) is slightly more complicated. Again we write $(1 - \phi_1 B - \phi_2 B^2)Y_t = \epsilon_t$, where by stationarity and causality the roots of the polynomial $\Phi(z) = 1 - \phi_1 z - \phi_2 z^2$ lie outside of the unit circle, that is $|z| > 1$. We denote the two roots by g_1^{-1} and g_2^{-1} . We may then express the polynomial as $\Phi(z) = (1 - g_1 z)(1 - g_2 z)$.

Then it follows that

$$Y_t = \frac{1}{(1 - g_1 B)(1 - g_2 B)} \epsilon_t.$$

Manipulating the double series we see

$$\begin{aligned} Y_t &= \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} g_1^{k_1} g_2^{k_2} B^{k_1+k_2} \epsilon_t \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} g_1^{k_1} g_2^{k_2} \epsilon_{t-(k_1+k_2)} \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n g_1^k g_2^{n-k} \epsilon_{t-n} \quad (n = k_1 + k_2) \\ &= \sum_{n=0}^{\infty} \epsilon_{t-n} \sum_{k=0}^n g_1^k g_2^{n-k}. \end{aligned}$$

In the third line above we use the fact that both $|g_1|, |g_2| < 1$, and hence we can apply the Cauchy Product formula for Series.

As a sum of a geometric series see that for $g_1 \neq g_2$

$$\sum_{k=0}^n g_1^k g_2^{n-k} = g_1^n \frac{1 - \frac{g_2^{n+1}}{g_1^{n+1}}}{1 - \frac{g_2}{g_1}} = \frac{g_1^{n+1} - g_2^{n+1}}{g_1 - g_2}.$$

And thus we can write

$$Y_t = \frac{g_1}{g_1 - g_2} \sum_{n=0}^{\infty} g_1^n \epsilon_{t-n} - \frac{g_2}{g_1 - g_2} \sum_{n=0}^{\infty} g_2^n \epsilon_{t-n}.$$

One can then substitute $g_1^{-1} = \frac{-\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2}$, $g_2^{-1} = \frac{-\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2}$.

When $g_1 = g_2$, we have $\sum_{k=0}^n g_1^k g_2^{n-k} = (n+1)g_1^n$.

Exercise 3.3

Assume we study an AR(2) process given by (where ϵ_t is a Gaussian process):

$$\frac{15}{16} Y_t = \frac{1}{4} Y_{t-1} - \frac{1}{16} Y_{t-2} + \epsilon_t, \quad t = 0, 1, 2, 3 \dots$$

Write down the characteristic AR polynomial associated with this process. Is this a stationary process? Is it invertible?

Solution 3.3

Rewriting our AR(2) process in terms of the backshift operator $\Phi(B)Y_t = \epsilon_t$ we get:

$$\left(\frac{15}{16} - \frac{1}{4}B + \frac{1}{16}B^2\right)Y_t = \epsilon_t. \quad (1)$$

We know that for the AR process to be stationary the roots of the characteristic polynomial $\Phi(z)$ must lie outside of the unit circle. We find out that the roots of the characteristic polynomial given in (1) are $2+3.3166247903554i$ and $2-3.3166247903554i$, respectively, which both lie outside the unit circle and hence our process is stationary. An AR process is always invertible.

Exercise 3.4

Assume we study the ARMA(2, 1) process generated according to

$$Y_t - \frac{1}{4}Y_{t-1} + \frac{1}{16}Y_{t-2} = \epsilon_t - 0.5\epsilon_{t-1}, \quad t = 0, 1, 2, 3, \dots$$

Is this a stationary process? Is it invertible?

Solution 3.4

The ARMA(2, 1) process is defined as $Y_t - \frac{1}{4}Y_{t-1} + \frac{1}{16}Y_{t-2} = \epsilon_t - 0.5\epsilon_{t-1}$ for $t \in \mathbb{N}$. This can be written in terms of the backshift operator as $(1 - \frac{1}{4}B + \frac{1}{16}B^2)Y_t = (1 - 0.5B)\epsilon_t$, giving characteristic polynomials $\Phi(z) = z^2 - 4z + 16$ and $\Omega(z) = 1 - \frac{1}{2}z$ for which we get respectively roots $z = 2 \pm 2\sqrt{3}i$ and $z = 2$ which are both outside the unit circle. Thus the process is stationary and invertible.

Exercise 3.5

Let $\{U_t\}$ be a stationary zero-mean time series. Define

$$X_t = (1 - 0.4B)U_t$$

and

$$W_t = (1 - 2.5B)U_t.$$

- (a) Express the autocorrelation functions of $\{X_t\}$ and $\{W_t\}$ in terms of that for $\{U_t\}$.
- (b) Show that $\{X_t\}$ and $\{W_t\}$ have the same autocorrelation functions.
- (c) Show that the process $\varepsilon_t = -\sum_{j=1}^{\infty} (0.4)^j X_{t+j}$ satisfies the difference equations

$$\varepsilon_t - 2.5\varepsilon_{t-1} = X_t.$$

Solution 3.5

1. We have $X_t = U_t - 0.4U_{t-1}$. Then for all $t, \tau \in T$ the autocovariance function is

$$\gamma_X(\tau) = \text{Cov}(U_t - 0.4U_{t-1}, U_{t+\tau} - 0.4U_{t+\tau-1}) = 1.16\gamma_U(\tau) - 0.4\gamma_U(\tau-1) - 0.4\gamma_U(\tau+1)$$

so the corresponding autocorrelation function is

$$\rho_X(\tau) = \frac{\gamma_X(\tau)}{\gamma_X(0)} = \frac{1}{\gamma_X(0)} \{1.16\gamma_U(\tau) - 0.4\gamma_U(\tau-1) - 0.4\gamma_U(\tau+1)\}$$

Remember that $\gamma_U(\tau) = \rho_U(\tau)\gamma_U(0)$, so

$$\rho_X(\tau) = \frac{\gamma_U(0)}{\gamma_X(0)} \{1.16\rho_U(\tau) - 0.4\rho_U(\tau-1) - 0.4\rho_U(\tau+1)\}$$

Using the fact that the autocovariance function is symmetric, we have $\gamma_U(-1) = \gamma_U(1)$, so

$$\gamma_X(0) = 1.16\gamma_U(0) - 0.8\gamma_U(1) = \gamma_U(0) \left\{ 1.16 - 0.8 \frac{\gamma_U(1)}{\gamma_U(0)} \right\} = \gamma_U(0) \{1.16 - 0.8\rho_U(1)\}$$

thus

$$\frac{\gamma_U(0)}{\gamma_X(0)} = \frac{1}{1.16 - 0.8\rho_U(1)}$$

Finally

$$\rho_X(\tau) = \frac{1}{1.16 - 0.8\rho_U(1)} \{1.16\rho_U(\tau) - 0.4\rho_U(\tau - 1) - 0.4\rho_U(\tau + 1)\}.$$

Similarly, we obtain

$$\rho_W(\tau) = \frac{1}{7.25 - 5\rho_U(1)} \{7.25\rho_U(\tau) - 2.5\rho_U(\tau - 1) - 2.5\rho_U(\tau + 1)\}$$

2. We note that

$$\rho_W(\tau) = \frac{6.25}{6.25 \{1.16 - 0.8\rho_U(1)\}} \{1.16\rho_U(\tau) - 0.4\rho_U(\tau - 1) - 0.4\rho_U(\tau + 1)\} = \rho_X(\tau)$$

3. Direct calculation shows that if $\varepsilon_t = -\sum_{j=1}^{\infty} 0.4^j X_{t+j}$ then

$$\begin{aligned} \varepsilon_t - 2.5\varepsilon_{t-1} &= -\sum_{j=1}^{\infty} 0.4^j X_{t+j} + 2.5 \sum_{j=1}^{\infty} 0.4^j X_{t+j-1} \\ &= -\sum_{j=1}^{\infty} 0.4^j X_{t+j} + 2.5 \sum_{l=0}^{\infty} 0.4^{l+1} X_{t+l} \\ &= -\sum_{j=1}^{\infty} 0.4^j X_{t+j} + X_t + \sum_{l=1}^{\infty} 0.4^l X_{t+l} \\ &= X_t \end{aligned}$$